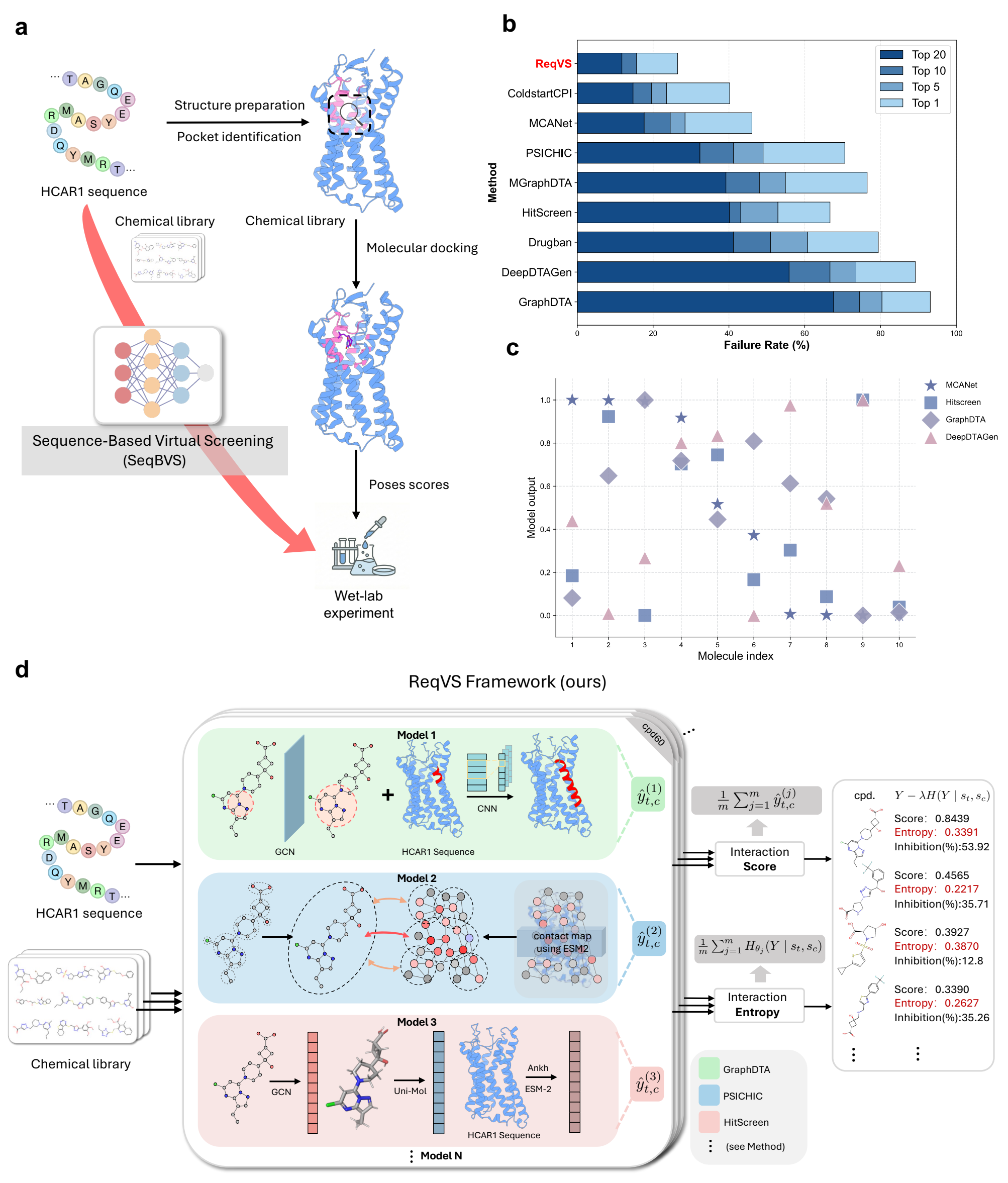
a
HCAR1 sequence
Structure preparation
Pocket identification
Chemical library
Chemical library
Molecular docking
Sequence-Based Virtual Screening (SeqBVS)
Poses scores
Wet-lab experiment
b
ReqVS
ColdstartCPI
MCANet
PSICHIC
MGraphDTA
HitScreen
Drugban
DeepDTAGen
GraphDTA
Method
Top 20
Top 10
Top 5
Top 1
Failure Rate (%)
c
MCANet
Hitscreen
GraphDTA
DeepDTAGen
Model output
Molecule index
d
ReqVS Framework (ours)
HCAR1 sequence
Chemical library
Model 1
GCN
HCAR1 Sequence
CNN
Model 2
contact map using ESM2
Model 3
GCN
Uni-Mol
Ankh
ESM-2
HCAR1 Sequence
Model N
cpd60
Interaction Score
Interaction Entropy
cpd.
Score: 0.8439
Entropy: 0.3391
Inhibition(%):53.92
Score: 0.4565
Entropy: 0.2217
Inhibition(%):35.71
Score: 0.3927
Entropy: 0.3870
Inhibition(%):12.8
Score: 0.3390
Entropy: 0.2627
Inhibition(%):35.26
GraphDTA
PSICHIC
HitScreen
(see Method)